

Proton Gradient Bench

Separating the two machines: a chain that pumps protons, and a synthase that spends them.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/proton-gradient-bench/

1 Before you open anything

Answer from what you already know. The point is to have a committed answer on paper before the bench gives you one.

PREDICTION

Oligomycin plugs the proton channel of ATP synthase. It binds nothing else — complexes I, II, III and IV are untouched. A mitochondrion respiring normally is dosed with it.

What happens to **oxygen consumption**? Circle one and give a number if you can: **rises / falls / unchanged**

Write one or two sentences saying what reasoning got you there.

2 Reading the resting bench

Open the demonstration. Nothing is added yet: the donor is NADH, substrate supply is 100%, oxygen is about 200 $\mu\text{mol L}^{-1}$.

1. Press **Reset everything**. Watch the recorder trace and wait until the three channels flatten out — a few seconds.
2. Fill in the table from the **Live readouts**, **Where the force actually lives** and **Energy budget** panels.
3. Read the note at the bottom of the **Energy budget** panel and copy the three proton fluxes it quotes into question c.

Readout	Value	Readout	Value
Electron flow		$\Delta\psi$ membrane potential	
O ₂ consumption		ΔpH across membrane	
Proton-motive force		Matrix pH	
ATP output		Intermembrane space pH	
P/O ratio		Heat released	

a. Divide *Electron flow* by *O₂ consumption*. You should get a whole number. Write the balanced half-reaction at complex IV that explains it.

b. The tiles report magnitudes, so the two components of the force add. Check that $\Delta\psi + (59.2 \text{ mV} \times \Delta\text{pH})$ gives the proton-motive force you recorded, and check that Matrix pH - Intermembrane space pH gives your ΔpH . Show both.

c. From the note: protons pumped out _____, protons returning through ATP synthase _____, protons leaking _____ (all nmol H⁺ min⁻¹ mg⁻¹). Divide the synthase figure by 4 and compare it with the ATP output you recorded. What does the 4 represent?

The 59.2 mV per pH unit is $2.303 RT/F$ **at 25 °C**. It is not a universal constant — at 37 °C the same ΔpH is worth about 61.5 mV. Every number on the bench is quoted at 25 °C.

3 What sets the speed of the chain

2,4-dinitrophenol carries protons across the membrane and does nothing else. It inhibits no protein. Use it as a dial on the gradient: more DNP, smaller gradient.

1. Press **Reset everything** and let it settle. That is your 0 row.
2. Drag the **Uncoupler (DNP)** slider to **10 $\mu\text{mol L}^{-1}$** , wait for the trace to flatten, and record the row.
3. Repeat for each dose in the table. The slider moves in steps of 1, so every value is reachable.

[DNP] $\mu\text{mol L}^{-1}$	Proton-motive force (mV)	O ₂ consumption	ATP output	Heat released
0				
10				
20				
30				
40				
100				

a. Two columns move in opposite directions as the dose rises. Name them, and complete the rule in your own words: *as the proton-motive force falls, oxygen consumption* _____, *because* _____.

b. At 100 $\mu\text{mol L}^{-1}$ the chain is running faster than it ever did with the synthase working, and ATP output is zero. Where is all that energy going? Quote the readout that answers it.

c. Now run your rule backwards. Set DNP back to **0**, then press **Oligomycin**. The synthase can no longer take protons back, so the force *rises*. Before you look: predict oxygen consumption. Then record what the bench gives.

Predicted _____ pmf _____ mV Measured O₂ consumption

d. Go back to your prediction in part 1. Was it right? If it was not, say specifically which step in your reasoning the data contradicts.

Nothing in the model treats oligomycin as an inhibitor of the chain. The complexes are untouched; electron flow is not blocked, it is **backed up** against a gradient the pumps themselves built. Keep that distinction in the wording of your answer.

4 Working backwards from a signature

Every drug on this bench drives ATP output to zero. Only the direction of the other two numbers says which machine was hit.

1. Press **Reset everything**, then work through **Rotenone**, **Cyanide** and **Oligomycin**, resetting between each. Note which way pmf and O₂ consumption move.
2. Now fill the table below *from your notes*, not by hunting: each row is an unlabelled result from a colleague's cuvette. Name the treatment that produces it.

	pmf	O ₂ consumption	ATP output	Treatment
A	220 mV, held	18	0	
B	falling to 0	0	0	
C	about 40 mV	about 130	0	

a. Row B is consistent with two different treatments. Name both, then describe the single change you would make on this bench to tell them apart, and what each outcome would mean.

b. A shortcut worth testing: *faster oxygen consumption means more ATP*. Reset, then switch the electron donor to **FADH₂ → complex II** and let it settle. Record both donors.

Donor	O ₂ consumption	ATP output	pmf	P/O ratio
NADH → complex I				
FADH ₂ → complex II				

c. Succinate makes the mitochondrion consume *more* oxygen and produce *less* ATP. Explain both halves of that in terms of protons per electron pair, and say in one sentence why the shortcut in **b** fails.

d. With FADH_2 still selected, press **Rotenone**. Predict first, then record: O_2 consumption

_____, ATP output _____. Account for the result using the position of the block.

5 Solve these without the demonstration

Close the page or look away. Show working. Then reopen the bench and check yourself — and where you disagree with it, find the step where you diverged.

1. The NADH route pumps 10 H^+ per electron pair (4 at I, 4 at III, 2 at IV); exporting one ATP costs 4 H^+ (3 through the rotor, 1 for the ADP/ATP and phosphate carriers). Calculate the theoretical P/O ceiling for NADH and for FADH_2 . Then, given that the resting bench pumps $1439 \text{ nmol H}^+ \text{ min}^{-1} \text{ mg}^{-1}$ and only 1324 of them return through the synthase, calculate the P/O you would actually expect to measure.

2. A mitochondrion at 25°C has a membrane potential of magnitude 172 mV, matrix pH 8.01 and intermembrane space pH 7.20. Calculate ΔpH , its value in millivolts, and the total proton-motive force. Which condition on the bench does this describe?

3. A mitochondrion carries a mutation that makes the inner membrane markedly more permeable to protons. No inhibitor is present and fuel is plentiful. State the direction of change — up, down or unchanged — for pmf, oxygen consumption, ATP output, heat released and P/O ratio, and **explain** why oxygen consumption moves the way it does even though nothing has been done to any complex. Which of the four treatments does this phenotype imitate?

4. On succinate, an oxygen electrode reports an electron flow of $371 \text{ nmol e}^- \text{ min}^{-1} \text{ mg}^{-1}$. Calculate (i) the oxygen consumption rate, (ii) the rate at which protons are pumped out, and (iii) the ATP output you would get if every one of those protons returned through the synthase. The bench actually reports $259 \text{ nmol ATP min}^{-1} \text{ mg}^{-1}$. Account for the difference, and check that your two numbers stand in the same ratio as the measured P/O does to its ceiling.

6 On your own

Nothing here is graded, and the worksheet has left most of the bench untouched.

- The *Substrate supply* slider was never used above. Take it from 100% to 50% to 10% and watch the P/O ratio rather than the rates. Why does starving the chain of fuel change how fast it runs without changing how efficiently it runs?
- The *Oxygen* slider is logarithmic, 0.01 to $250 \mu\text{mol L}^{-1}$, and complex IV has a K_m of $0.5 \mu\text{mol L}^{-1}$. Find the concentration at which oxygen consumption has dropped to about half its resting value, and compare it with the K_m . How far do you have to drag the slider before anything at all happens, and what does that imply about tissue at 5 to $30 \mu\text{mol L}^{-1}$?
- Add cyanide and watch the recorder trace rather than the tiles. ATP output does not reach zero the instant the chain stops. Read the pmf channel over the same seconds and say what is paying for that last ATP.
- Read the *Honest limits* panel. It notes that a real ATP synthase can spin backwards, hydrolysing ATP to defend the gradient. Sketch what the three recorder channels would do after cyanide if

this model allowed that, and say which of your answers above it would change.